

PYTHON PROGRAMMING MINI PROJECT

Personal Expense Tracker

A Command-Line Application for Personal Finance Management

Submitted by

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VaultofCodes Python Internship

Assignment 3 – Week 3

1. Project Objective

The Personal Expense Tracker is a command-line Python application developed as part of the VaultofCodes Python Programming Internship, Assignment 3 (Week 3). The primary objective of this project is to build a practical, menu-driven tool that allows a user to record and manage daily expenses in an organized and efficient manner.

The project is designed to allow users to record their daily expenses, organize those expenses into predefined categories, and view meaningful spending summaries at both the category and monthly level. It also enables users to save data persistently using JSON file handling, so that expense records remain available across multiple program sessions. In addition, the application allows users to analyze their spending patterns using graphical visualization powered by Matplotlib.

Overall, the project reinforces core Python programming concepts such as functions, lists, dictionaries, file handling, input validation, and data visualization, while producing a genuinely usable expense-tracking tool.

2. Technologies Used

The following technologies and programming concepts were used to build this project:

- **Python 3** — Used as the core programming language for building the entire application logic, menu system, and control flow.
- **Lists and Dictionaries** — Used to store multiple expense records and to represent each individual expense as a structured record with fields such as amount, category, description, date, and time.
- **JSON File Handling** — Used to persistently store and retrieve expense data between program runs, using Python's built-in json module to read from and write to an expenses.json file.
- **Matplotlib** — Used to generate a bar chart that visually represents total spending across different expense categories, helping the user analyze spending patterns at a glance.
- **Functions and Modular Programming** — The entire application is organized into well-defined functions, with each feature implemented as an independent, reusable function. This modular approach improves code readability, maintainability, and testing.

3. Project Features

The Personal Expense Tracker implements the following features:

- **Add Expense** — Allows the user to record a new expense by entering the amount, selecting a category, adding a description, and choosing a date and time.
- **View All Expenses** — Displays every recorded expense in a formatted tabular layout, including ID, amount, category, description, date, and time.
- **Category-wise Expense Summary** — Calculates and displays the total amount spent in each expense category.
- **Overall Spending Summary** — Displays the total number of expense records along with the grand total amount spent.

- **Monthly Spending Summary** — Groups and displays total expenses on a month-by-month basis, based on the date of each entry.
- **Search Expense by Category** — Allows the user to filter and view all expenses belonging to a specific category, along with the category subtotal.
- **Edit Expense** — Allows the user to locate an expense by its ID and update its amount, category, description, or date, while retaining the existing value for any field left unchanged.
- **Delete Expense** — Allows the user to locate an expense by its ID, review its details, and remove it after confirmation.
- **Matplotlib Expense Analytics Chart** — Generates a bar chart of total spending by category using Matplotlib, providing a visual summary of spending distribution.
- **JSON Data Persistence** — Automatically saves all expense records to an expenses.json file and reloads them the next time the program is run, ensuring no data is lost between sessions.
- **Input Validation and Error Handling** — Validates user input throughout the program (such as amounts, category choices, and dates) and handles invalid entries gracefully with clear error messages, preventing the program from crashing on incorrect input.

4. Implementation and Code Explanation

This section briefly explains the purpose of each major function used in the project.

add_expense() Collects the expense amount, category, description, and date/time from the user, validates the input, creates a new expense record, and saves it to the expense list and JSON file.

view_expenses() Displays all recorded expenses in a formatted table, showing the ID, amount, category, description, date, and time for each entry.

category_summary() Iterates through all expenses and calculates the total amount spent in each category, then displays the results.

overall_summary() Calculates and displays the total number of expenses recorded and the overall (grand total) amount spent.

monthly_summary() Groups expenses by month (extracted from the date field) and displays the total amount spent in each month.

search_by_category() Allows the user to select a category and displays only the expenses that belong to that category, along with the category subtotal.

edit_expense() Locates an expense by its ID and allows the user to update its amount, category, description, or date. Any field left blank retains its existing value.

delete_expense() Locates an expense by its ID, displays its details for confirmation, and removes it from the records upon user confirmation.

show_expense_chart() Calculates the total spending per category and uses Matplotlib to display a bar chart summarizing the expense distribution.

save_expenses() Writes the current list of expense records to the expenses.json file, ensuring that all changes are stored persistently.

load_expenses() Reads and loads previously saved expense records from the expenses.json file when the program starts. If the file does not exist or is corrupted, it safely initializes an empty expense list.

Each expense is represented as a dictionary containing keys such as id, amount, category, description, date, and time, and all such dictionaries are stored together inside a single list. This list is saved persistently in the expenses.json file using JSON file handling, so that the data remains available even after the program is closed. When the program starts, it automatically loads any previously saved data from this file, allowing the user to continue managing their expense records across multiple sessions.

5. User Menu

The application presents the following main menu to the user each time it runs:

1	Add Expense
2	View All Expenses
3	View Category Summary
4	View Overall Summary
5	View Monthly Summary
6	Search Expense by Category
7	Edit Expense
8	Delete Expense
9	Show Expense Chart
10	Exit

The user selects an option by entering the corresponding number (1–10), and the program executes the associated feature before returning to the main menu.

6. Output and Testing

The application was tested by running it through each menu option and verifying that the expected output was produced. The following screenshots demonstrate the working of the program:

Screenshot 1: Main Menu

```
PS C:\Users\kanak\OneDrive\Desktop\VaultofCodes-Python-Internship\Assignment-3> cd Task-1_Personal-Expense-Tracker
PS C:\Users\kanak\OneDrive\Desktop\VaultofCodes-Python-Internship\Assignment-3\Task-1_Personal-Expense-Tracker> python expense_tracker.py

=====
          PERSONAL EXPENSE TRACKER
=====
1. Add Expense
2. View All Expenses
3. View Category Summary
4. View Overall Summary
5. View Monthly Summary
6. Search Expense by Category
7. Edit Expense
8. Delete Expense
9. Show Expense Chart
10. Exit
=====
Enter your choice (1-10):
```

Screenshot 2: Adding An Expense

```
Enter your choice (1-10): 1

=====
          PERSONAL EXPENSE TRACKER
=====
Add New Expense

Enter Amount (₹): 10

Available Categories:

1. Food
2. Transport
3. Shopping
4. Entertainment
5. Bills
6. Health
7. Education
8. Other

Choose Category (1-8): 1
Enter Description: kitkat

Date Options
1. Use Today's Date & Time
2. Enter Custom Date & Time
Choose (1/2): 1

Expense added successfully!
Expense ID : 11
Amount      : ₹10.00
Category    : Food
Date        : 2026-07-24
Time        : 22:56:41

Press Enter to continue...
```

Screenshot 3: View All Expenses

```

=====
                PERSONAL EXPENSE TRACKER
=====
1. Add Expense
2. View All Expenses
3. View Category Summary
4. View Overall Summary
5. View Monthly Summary
6. Search Expense by Category
7. Edit Expense
8. Delete Expense
9. Show Expense Chart
10. Exit
=====
Enter your choice (1-10): 2

=====
                PERSONAL EXPENSE TRACKER
=====

```

ID	Amount	Category	Description	Date	Time
1	₹100.00	Food	hotel brijwasi centrum	2026-07-20	--:--:--
2	₹2.00	S	qs	2026-07-20	--:--:--
3	₹2.00	S	s	2026-07-20	--:--:--
4	₹1223.00	Svhdv	bhbs	2026-07-20	--:--:--
5	₹3234.00	1Ce	fre23	2026-07-20	--:--:--
6	₹20.00	Coffee	agrawal	2026-07-20	--:--:--
7	₹30.00	Food	choclalte	2026-07-20	--:--:--
8	₹10.00	Food	toffee	2026-07-20	19:16:09
9	₹10.00	Food	da	2026-07-20	19:17:10
10	₹12.00	Bills	15dsc	2026-07-24	22:43:49
11	₹10.00	Food	kitkat	2026-07-24	22:56:41

```

-----
Total Records : 11

```

Screenshot 4: Category / Overall / Monthly Summary

```
=====
                        PERSONAL EXPENSE TRACKER
=====
1. Add Expense
2. View All Expenses
3. View Category Summary
4. View Overall Summary
5. View Monthly Summary
6. Search Expense by Category
7. Edit Expense
8. Delete Expense
9. Show Expense Chart
10. Exit
=====
Enter your choice (1-10): 3
=====
                        PERSONAL EXPENSE TRACKER
=====
Category-wise Expense Summary

Category                Total Amount
-----
Food                    ₹160.00
S                        ₹4.00
Svhdv                   ₹1223.00
1Ce                      ₹3234.00
Coffee                  ₹20.00
Bills                   ₹12.00

Press Enter to continue...
```



```
=====
                PERSONAL EXPENSE TRACKER
=====
1. Add Expense
2. View All Expenses
3. View Category Summary
4. View Overall Summary
5. View Monthly Summary
6. Search Expense by Category
7. Edit Expense
8. Delete Expense
9. Show Expense Chart
10. Exit
=====
Enter your choice (1-10): 4

=====
                PERSONAL EXPENSE TRACKER
=====
Overall Expense Summary

Total Expenses : 11
Grand Total    : ₹4653.00

Press Enter to continue...
```

```
=====
                PERSONAL EXPENSE TRACKER
=====
1. Add Expense
2. View All Expenses
3. View Category Summary
4. View Overall Summary
5. View Monthly Summary
6. Search Expense by Category
7. Edit Expense
8. Delete Expense
9. Show Expense Chart
10. Exit
=====
Enter your choice (1-10): 5
=====
                PERSONAL EXPENSE TRACKER
=====
Monthly Expense Summary

Month          Amount
-----
2026-07        ₹4653.00
```

Screenshot 5: Search, Edit or Delete Expense

```
=====
                PERSONAL EXPENSE TRACKER
=====
1. Add Expense
2. View All Expenses
3. View Category Summary
4. View Overall Summary
5. View Monthly Summary
6. Search Expense by Category
7. Edit Expense
8. Delete Expense
9. Show Expense Chart
10. Exit
=====
Enter your choice (1-10): 5

=====
                PERSONAL EXPENSE TRACKER
=====
Monthly Expense Summary

Month          Amount
-----
2026-07        ₹4653.00
```

```

PERSONAL EXPENSE TRACKER
=====
Monthly Expense Summary

Month          Amount
-----
2026-07        ₹4653.00

Press Enter to continue...6

PERSONAL EXPENSE TRACKER
=====
1. Add Expense
2. View All Expenses
3. View Category Summary
4. View Overall Summary
5. View Monthly Summary
6. Search Expense by Category
7. Edit Expense
8. Delete Expense
9. Show Expense Chart
10. Exit
=====
Enter your choice (1-10): 7

PERSONAL EXPENSE TRACKER
=====
Available Expenses:

ID: 1 | ₹100.00 | Food | hotel brijwasi centrum | 2026-07-20
ID: 2 | ₹2.00 | S | qs | 2026-07-20
ID: 3 | ₹2.00 | S | s | 2026-07-20
ID: 4 | ₹1223.00 | Svhdv | bhbs | 2026-07-20
ID: 5 | ₹3234.00 | 1Ce | fre23 | 2026-07-20

```

```
=====
PERSONAL EXPENSE TRACKER
=====
Available Expenses:

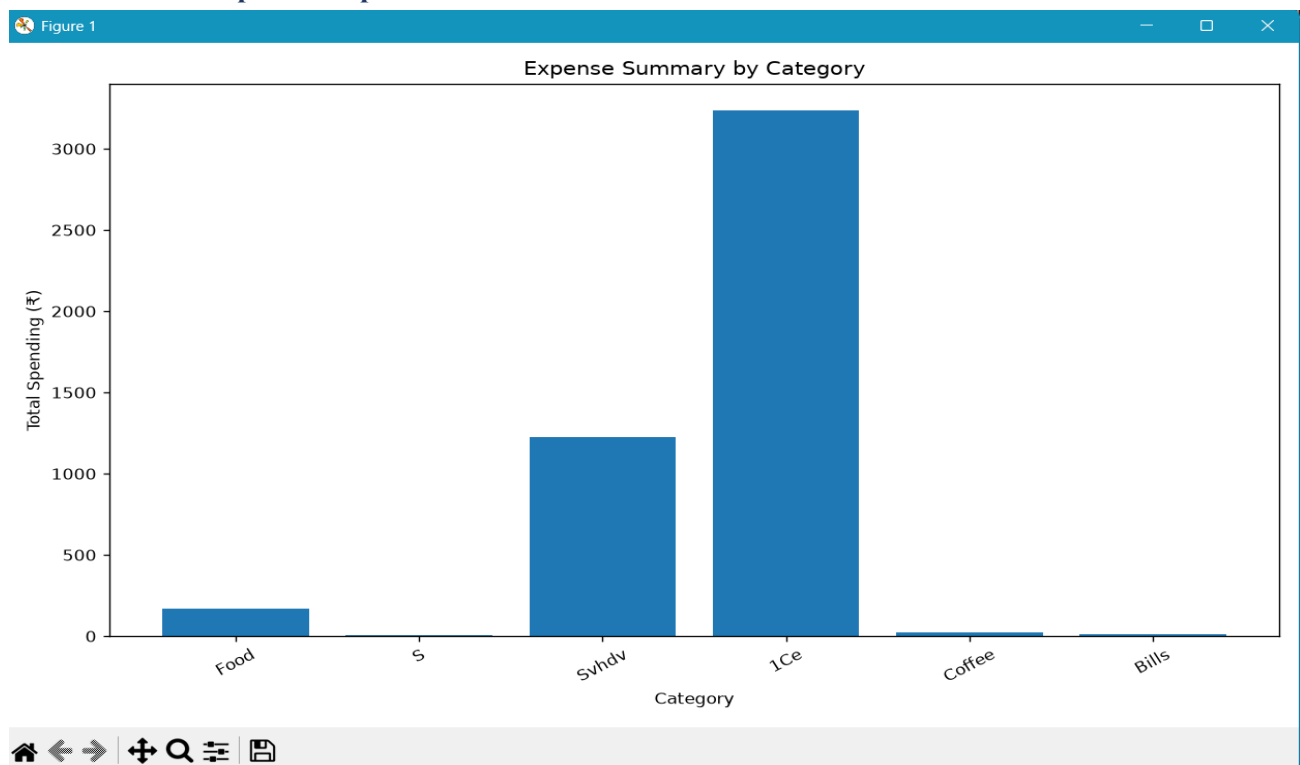
ID: 1 | ₹100.00 | Food | hotel brijwasi centrum | 2026-07-20
ID: 2 | ₹2.00 | S | qs | 2026-07-20
ID: 3 | ₹2.00 | S | s | 2026-07-20
ID: 4 | ₹1223.00 | Svhdv | bhbs | 2026-07-20
ID: 5 | ₹3234.00 | 1Ce | fre23 | 2026-07-20
ID: 6 | ₹20.00 | Coffee | agrawal | 2026-07-20
ID: 7 | ₹30.00 | Food | choclate | 2026-07-20
ID: 8 | ₹10.00 | Food | toffee | 2026-07-20
ID: 9 | ₹10.00 | Food | da | 2026-07-20
ID: 10 | ₹12.00 | Bills | 15dsc | 2026-07-24
ID: 11 | ₹10.00 | Food | kitkat | 2026-07-24

Enter Expense ID to edit: 8

Press Enter to keep the existing value.

Enter new amount (Current: ₹10.00):
```

Screenshot 6: Matplotlib Expense Chart



7. Project Structure

The project files are organized as follows:

```
Assignment-3
├── Task-1_Personal-Expense-Tracker
│   ├── expense_tracker.py
│   ├── expenses.json
│   └── Assignment-3_Report.pdf
```

8. Conclusion

The Personal Expense Tracker project successfully demonstrates a range of core Python programming concepts, including the use of functions and modular programming, lists and dictionaries for structured data storage, file handling for persistent data storage, input validation for robust error handling, and basic data visualization using Matplotlib.

By combining these concepts into a single, cohesive command-line application, the project provides a practical and genuinely useful solution for tracking and analyzing personal expenses. It allows a user to add, view, search, edit, and delete expense records, generate category-wise and monthly summaries, and visualize spending patterns through a bar chart — all while ensuring that data is safely stored and retrieved using JSON file handling.

This assignment strengthened practical skills in Python application development, including working with real-world data persistence and building a clean, user-friendly command-line interface.